

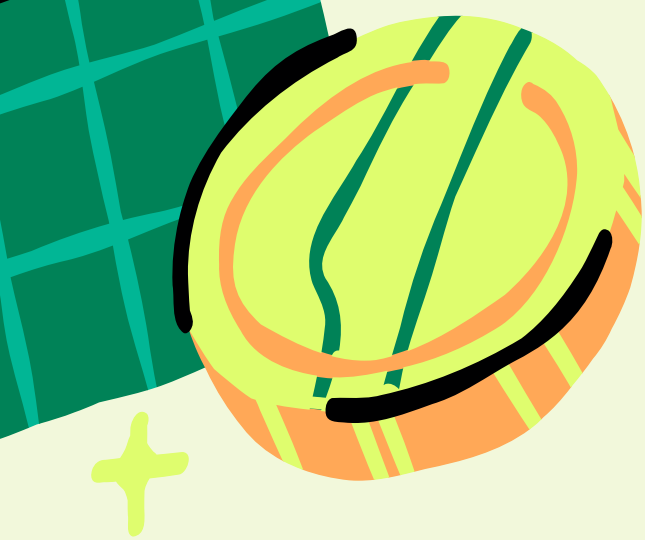
The background is a light cream color with various hand-drawn financial icons. In the top left, there's a large yellow coin with an orange dollar sign. To its right is another yellow coin with orange stripes. Further right is a stack of blue and yellow banknotes. In the top right corner, there's a green grid pattern and a yellow coin with orange stripes. In the bottom left, there are several yellow coins with orange dollar signs. In the bottom center, there's a stack of blue and yellow banknotes. In the bottom right, there's a yellow coin with orange stripes and a green and yellow abstract shape. Small yellow and orange starburst icons are scattered throughout the background.

CreditFair

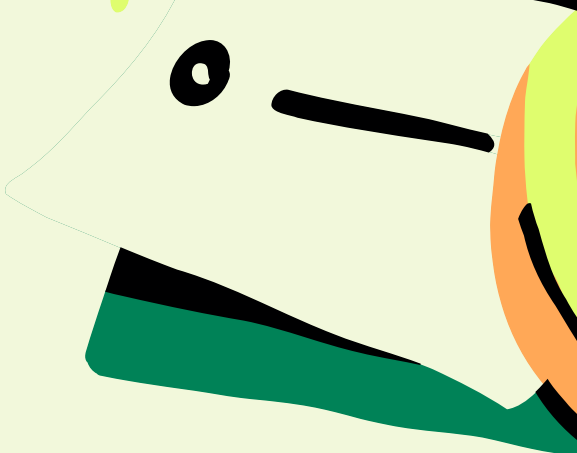
Our Team

Simran Amesar

Shreya Challagulla



Introduction



What it is: CreditFair predicts whether a loan applicant is a good or bad credit risk on the German Credit dataset.

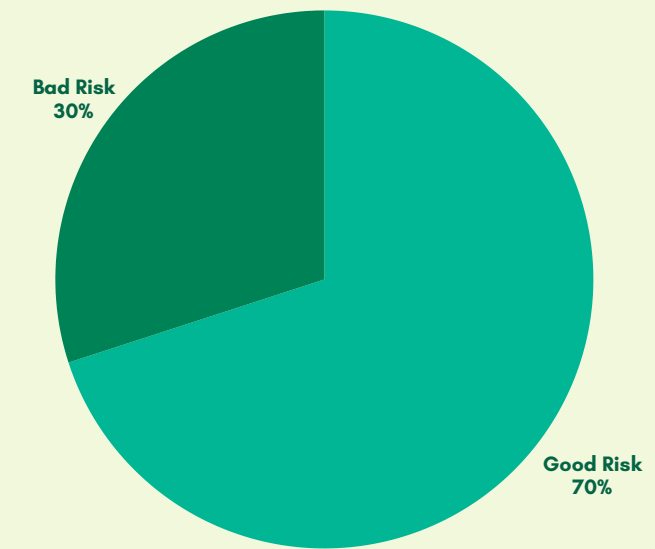
Why it matters: Built to be compliant, fair, explainable and audited — a high-risk AI system under the EU AI Act.

Dataset: German Credit Data (Hofmann, 1994, UCI) — full Statlog version: 1,000 applicants, 20 features, official 5:1 cost matrix.



Data Analysis

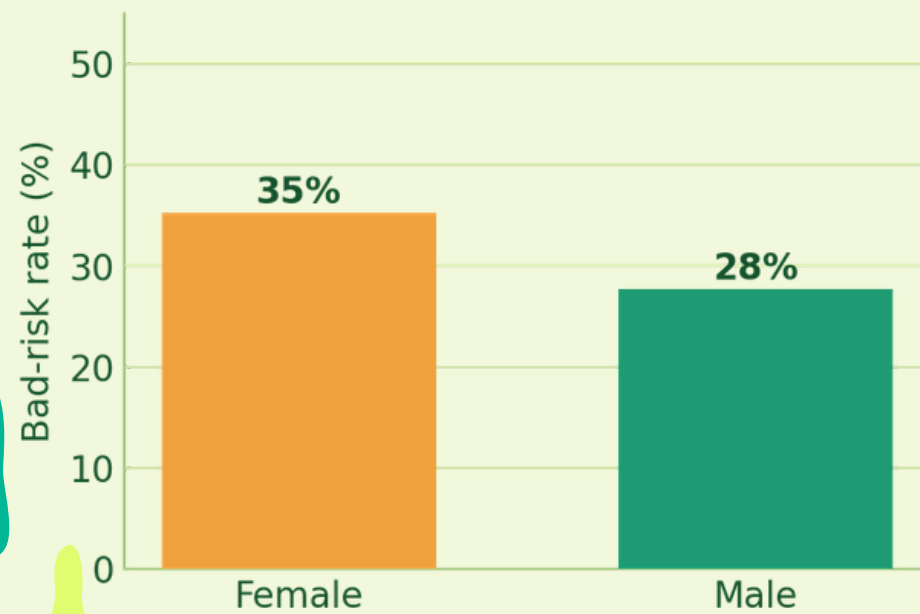
- **Size & target:** 1,000 applicants — 700 good / 300 bad (30% bad-risk).
- **No missing values:** the full encoding uses explicit categories (e.g. “no account”) instead of blanks.
- **Strongest predictor:** checking-account status — <0 DM $\rightarrow$ 49% bad vs no-account $\rightarrow$ 12%; loan duration next.
- **Cost asymmetry:** approving a bad applicant costs 5× a wrongful refusal (5:1 matrix).



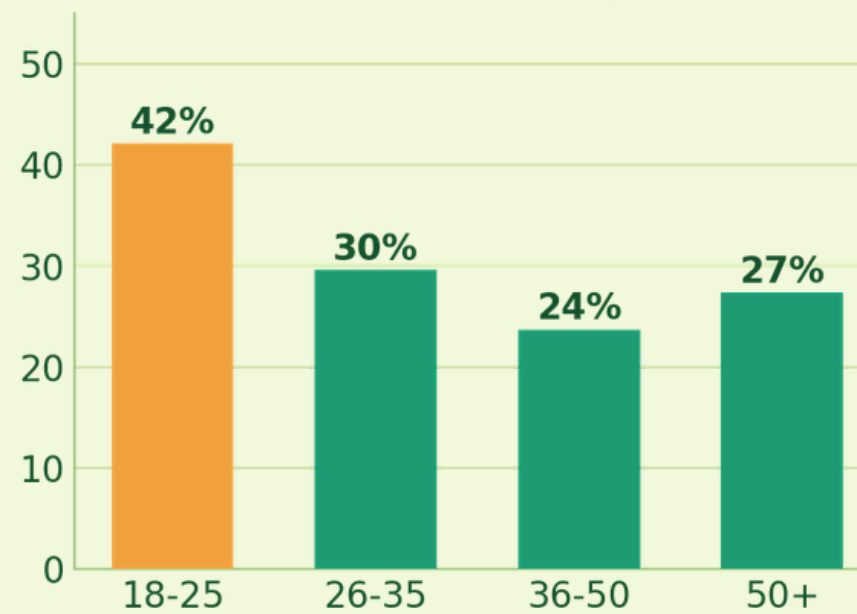
Fairness Gaps

Historical fairness gaps · 1,000 applicants · overall bad-risk 30% · proxy AUC: sex 0.70, foreign 0.82

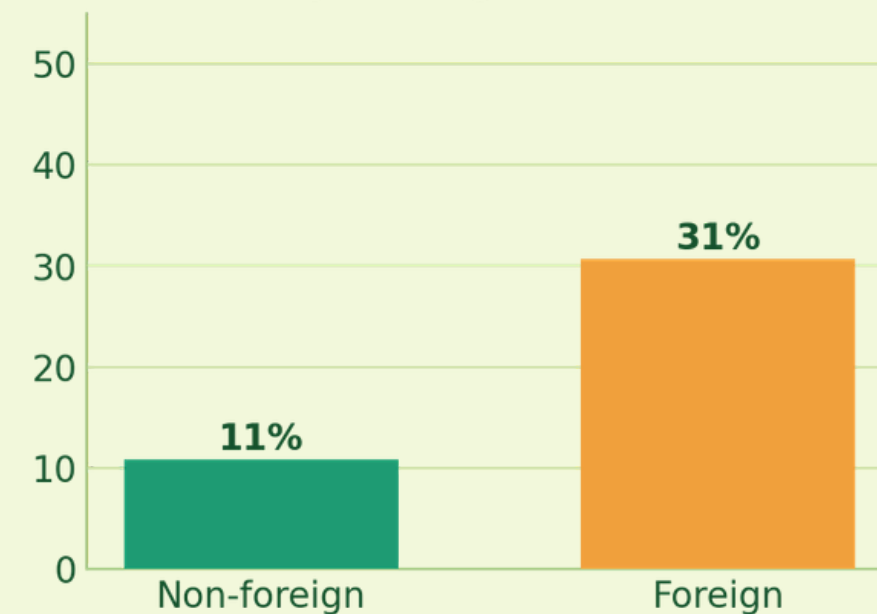
By Sex



By Age group



By Foreign worker



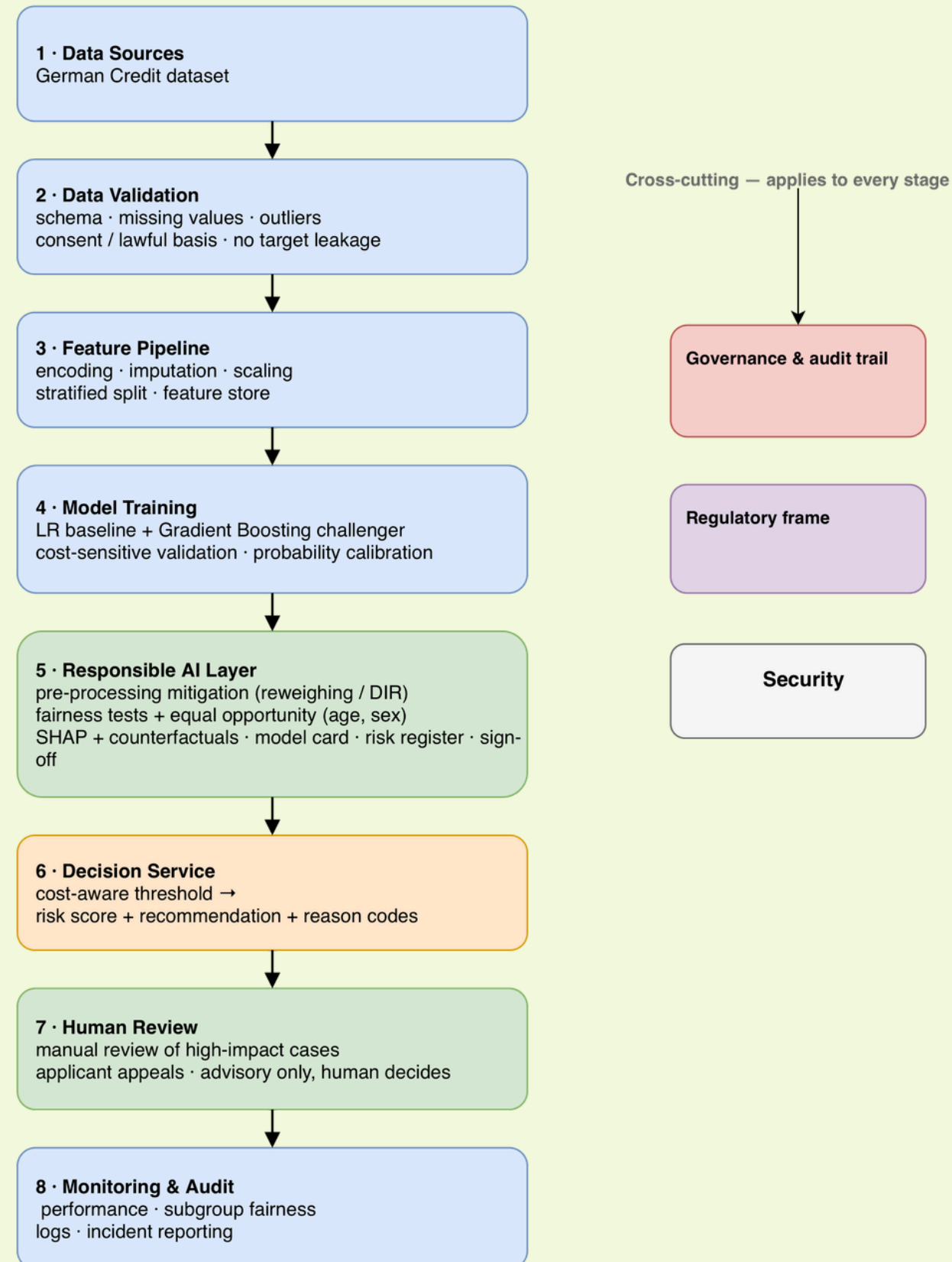
(non-foreign n=37 — small group)

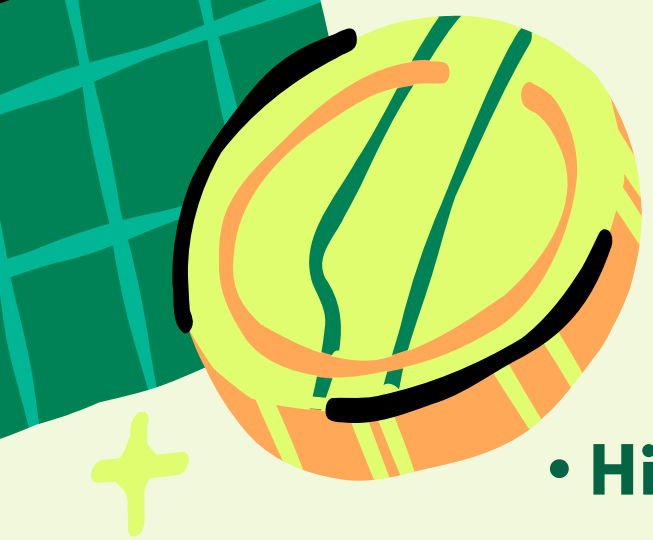
Plan

- **Metric:** expected cost (5:1) + ROC-AUC + recall on bad — not accuracy.
- **Imbalance:** class weights / resampling on the 700/300 split; stratified split.
- **Fairness:** measure across age, sex and foreign-worker; target equal opportunity + DI ≥ 0.80 .
- **Mitigation:** lead with pre-processing (reweighing / DIR) — not protected attributes in the live decision.
- **Scope:** in — baseline + fairness + XAI + tests + model card; out — deep nets, external data, UI.

Proposed Architecture

CreditFair — Responsible AI Credit Scoring Architecture





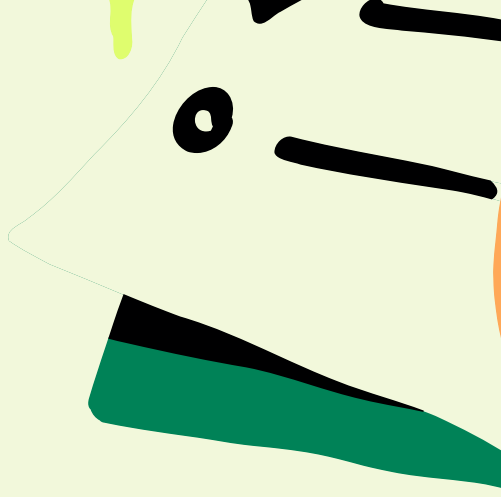
Regulatory analysis



- **High-risk AI:** credit scoring is high-risk under EU AI Act, Annex III 5(b); duties (Art. 8-15) apply 2 Aug 2026.
 - **GDPR Art. 22 + SCHUFA (C-634/21):** the score itself is an automated decision → human review, explanation & appeal are mandatory.
 - **German AGG:** no discrimination by age, sex or foreign-worker status; audit with the four-fifths rule ($DI \geq 0.80$).
 - **Before deployment:** DPIA + Fundamental Rights Impact Assessment (FRIA).
 - **Design answer:** advisory-only, human-in-the-loop, reason codes, fairness auditing, full audit log.
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Week 1 Status



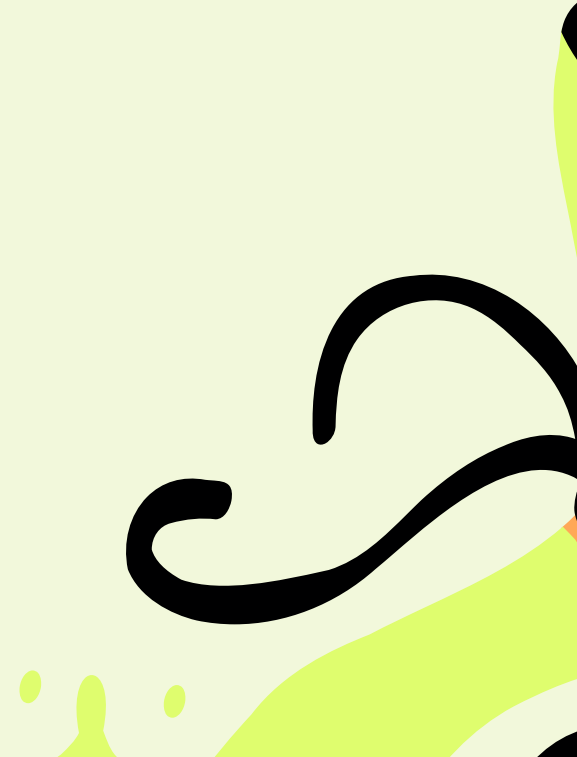
Done this week

- Loaded & decoded the 1,000-row Statlog data
 - Full EDA + fairness (age, sex, foreign)
 - Plan, architecture & regulatory analysis
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Challenges

- Decoding the coded A^{**} categories
 - Counter-intuitive credit-history pattern
 - Tiny non-foreign group ($n=37$) — fragile
 - Imbalance vs the 5:1 cost trade-off
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Next (Week 2)

- Baseline + gradient-boosting models
 - Pre-processing fairness mitigation
 - Fairness-accuracy trade-off
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Thank You